

Isobaric Vapor–Liquid Equilibrium Data for the Isopropanol–Water System

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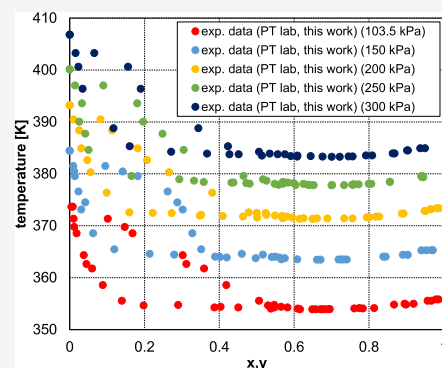
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ABSTRACT: This work presents experimental isobaric vapor–liquid equilibrium data for the isopropanol–water system collected at 150 kPa, 200 kPa, 250 kPa, and 300 kPa at the Process Thermodynamics laboratory (PT lab) of Politecnico di Milano and its representation with the thermodynamic models PSRK, NRTL, UNIFAC, and Wilson. The system is strongly nonideal, with the presence of a minimum boiling azeotrope which was found at all the pressures of analysis. Data at the considered pressures for the system are not available in the literature even though they are of interest for thermodynamic modeling of strongly nonideal mixtures which employ thermodynamic parameters regressed on the basis of experimental points. The reliability of the operation of the experimental unit and of the measurements was checked by determining the vapor pressure curve of the single components and by comparing it with the ones from the literature and also by comparing the collected data for the binary system with those available in the literature at atmospheric pressure. In addition, the test on thermodynamic consistency has been performed for all the collected data.



1. INTRODUCTION

The increasing greenhouse gas emissions and the resulting global climate change, joined with a growing world population,

Table 1. Experimental Vapor–Liquid Equilibrium Data for the Isopropanol–Water System at Atmospheric Pressure

source	no. of experimental points	$T_{\min}$ [K]	$T_{\max}$ [K]
Lebo (1921) ²²	42	353.65	373.15
Brunjes and Bogart (1943) ²⁵	27	353.03	371.87
Wilson and Simons (1952) ²³	26	353.11	373.00
Dobroserdov (1959) ²⁶	14	353.31	366.32
Udovenko and Mazanko (1967) ²⁷	18	353.53	369.66
Marzal et al. (1996) ²⁸	27	352.65	372.78
Khalifaoui et al. (1997) ²⁹	12	352.25	373.13
Sevgili and Senol (2006) ³⁰	17	353.16	373.15
Lin and Tu (2014) ³¹	17	353.22	373.15

are key challenges that are leading to an increased interest in processes able to produce fuels and chemicals from alternative, renewable resources. In this respect, the production of bioethanol, which can be used as raw material for ethylene production by its dehydration¹ or for hydrogen production by steam reforming,² can be mentioned. Also butanol is an important commodity chemical and an attractive biofuel, which can be produced through the acetone–butanol–ethanol

(ABE) fermentation by clostridia that is characterized by high process costs, due to the large volumes required and the downstream product recovery from dilute broths, usually performed by distillation.³ These processes involve mixtures containing alcohols and water, which make the downstream processing expensive⁴ and, for many mixtures containing alcohols, a potentially difficult separation challenge due to the formation of azeotropes. Another system of this type is the one involving isopropanol and water.⁵

Nowadays, isopropanol (2-propanol, also known as isopropyl alcohol, IPA), is a solvent and an intermediate for chemicals production. For example, it is used for the production of monoisopropylamine and isopropyl acetate both as a solvent and as an intermediate.⁶ IPA is commonly used as a solvent also in pharmaceutical and electronic industries. It is proposed as a substitute for hexane in vegetable oil extraction⁷ because of the advantages regarding safety, and it can be partially recovered from the extract only by cooling due to its partial miscibility with oil.

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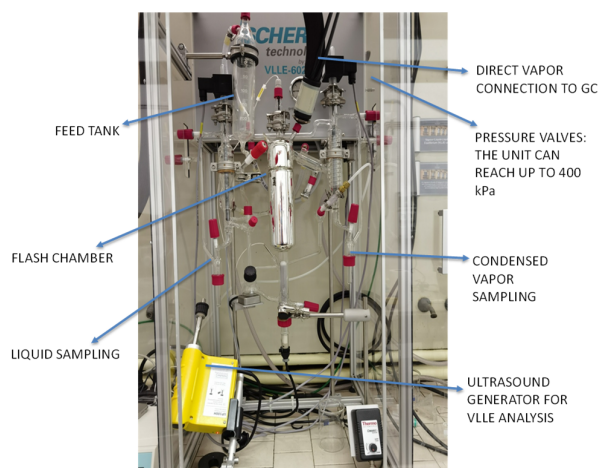


Table 2. Isobaric Experimental Vapor–Liquid Equilibrium Data for the Isopropanol–Water System at Pressures Different from 1 atm

source	pressure [kPa]	no. of experimental points	$T_{\min}$ [K]	$T_{\max}$ [K]
Wilson and Simons (1952) ²³	12.66	24	308.78	323.71
Wilson and Simons (1952) ²³	38.66	24	322.23	338.29
Wilson and Simons (1952) ²³	50.66	25	336.90	354.68
Wilson and Simons (1952) ²³	411.566	16	393.40	477.61
Marzal et al. (1996) ²⁸	30	25	325.33	342.33
Marzal et al. (1996) ²⁸	60	26	340.52	359.14

Table 3. Sample Description Table

chemical name	source	purity (mole fraction)	analysis method
2-propanol	Sigma-Aldrich	>99.8%	gas chromatography
demineralized water	Idrochimica Srl	ASTM D 1193 TYPE 4	gas chromatography
nitrogen	Sapio	>99.99%	gas chromatography

**Figure 1.** Unit for the study of VLE at the PT lab.

Isopropanol can be found in a mixture with water in several processes in which it is involved and in its production by indirect hydrogenation or by fermentation of cellulosic materials. To obtain pure IPA, the raw product stream needs to be dehydrated. Since the mixture IPA + water forms an azeotrope (at 101.3 kPa at about 87.5 wt %), the system is strongly nonideal and the separation into almost pure components is complex and energy-intensive. Isopropanol–water separation started to raise interest early in the last century and has been then studied in recent years to develop efficient methods to obtain pure isopropanol. Several processes have been suggested such as azeotropic distillation,^{5,8} pressure swing distillation,⁹ a single unit of pervaporation membrane or multistage arrangements for the dehydration of isopropanol under adiabatic and ideal isothermal conditions,¹⁰ liquid–liquid extraction,¹¹ reactive distillation,¹² adsorption,¹³ the adoption of ionic liquids for breaking the azeotrope,¹⁴ and extractive distillation through deep eutectic solvents, also known as low transition temperature mixtures (LTTMs).^{15–20}

The properties of the isopropanol–water mixture were studied in the 19th century as reported by Young in 1902,²¹ which determined some boiling points at atmospheric pressure. In 1921, Lebo²² investigated the specific gravity and proposed data of vapor–liquid equilibrium for the distillation, reporting that “the boiling point and composition of the vapor from any liquid

Table 4. Data of Vapor Pressure of Water Collected at the PT lab of Politecnico di Milano (a Maximum Oscillation of 0.03 K is Tolerated for Each Point)

pressure [kPa]	temp [K]
35.00	345.82
40.00	348.99
45.00	351.84
50.00	354.42
55.00	356.81
60.00	359.01
65.00	361.07
70.00	362.99
75.00	364.82
80.00	366.53
85.00	368.17
90.00	369.73
95.00	371.20
101.30	373.10
110.00	375.42
120.00	377.87
130.00	380.19
140.00	382.31
150.00	384.45
160.00	386.39
170.00	388.05
180.00	389.85
190.00	391.61
200.00	393.25
210.00	394.76
220.00	396.20
230.00	397.61
240.00	399.03
250.00	400.35
260.00	401.60
270.00	402.55
280.00	403.45
290.00	404.60
300.00	405.75
310.00	406.83
320.00	407.95
330.00	409.03
340.00	410.05
350.00	411.05

mixture depend upon the composition of the liquid”, therefore confirming the importance of vapor–liquid equilibrium data.

1.1. Isobaric Data Available in the Literature. To the authors’ knowledge, the works that propose vapor–liquid equilibrium (VLE) data in the last century mainly investigate the atmospheric conditions, with only a few works focusing on pressures different from 1 atm. In 1952, Wilson and Simons²³ investigated the azeotropic mixture at the pressure of 412 kPa.

Table 5. Data of Vapor Pressure of Isopropanol Collected at the PT lab of Politecnico di Milano (a Maximum Oscillation of 0.03 K Is Tolerated for Each Point)

pressure [kPa]	temp [K]
31.07	328.69
35.00	331.15
40.00	333.92
45.00	336.47
50.00	338.76
52.70	339.85
55.00	340.88
60.00	342.84
65.00	344.67
70.00	346.40
75.00	348.02
80.00	349.54
85.00	351.01
90.00	352.40
95.00	353.73
103.50	355.75
110.00	357.27
120.00	359.48
130.00	361.53
140.00	363.45
150.00	365.25
160.00	366.98
170.00	368.60
180.00	370.21
190.00	371.72
200.00	373.21
210.00	374.62
220.00	375.95
230.00	377.26
240.00	378.51
250.00	379.72
260.00	380.88
270.00	382.02
280.00	383.09
290.00	384.13
300.00	385.41
310.00	386.46
320.00	387.44
330.00	388.41
340.00	389.35
350.00	390.26

Barr-David and Dodge in 1959²⁴ published a study in which the pressure was investigated in the range of about 520–12350 kPa, though considering only isothermal vapor–liquid equilibrium curves. Table 1 and Table 2 report the isobaric data found in the literature for the considered system. No isobaric VLE data in the pressure range from 150 kPa to less than 400 kPa, which are the object of the study presented in this work, were found.

1.2. Aim of the Work. This work aims at collecting isobaric vapor–liquid equilibrium data for the isopropanol–water system at pressures intermediate between 101.3 and 400 kPa, for which no data can be found in the literature, in particular at 150 kPa, 200 kPa, 250 kPa, and 300 kPa. These data are fundamental for choosing the correct thermodynamic model to be used for the simulation of the processes in which isopropanol and water are present, both in chemical and in energy processes. Indeed, separation operations play a central role in the practice

Table 6. Data of Temperature, Pressure, Composition in the Liquid and in the Vapor Phases Collected at the PT lab of Politecnico di Milano at a Pressure of 103.5 kPa and Related Average Uncertainties (x Is the Mole Fraction in the Liquid Phase and y Is the Mole Fraction in the Vapor Phase of the Most Volatile Component, i.e., Isopropanol)^a

pressure [kPa]	temp [K]	x	y
103.5	373.69	0.0078	0.0042
103.5	371.36	0.0104	0.1018
103.5	369.78	0.0121	0.1470
103.5	368.57	0.0181	0.1679
103.5	364.34	0.0376	0.3018
103.5	362.63	0.0447	0.3118
103.5	361.75	0.0597	0.3591
103.5	358.57	0.0883	0.4185
103.5	355.56	0.1393	0.5062
103.5	354.65	0.1974	0.5303
103.5	354.76	0.2897	0.5467
103.5	354.26	0.3869	0.5473
103.5	354.41	0.4032	0.5653
103.5	354.27	0.4507	0.5819
103.5	354.05	0.5379	0.6099
103.5	353.92	0.6156	0.6487
103.5	353.90	0.6578	0.6703
103.5	353.91	0.6756	0.6837
103.5	353.94	0.6934	0.6951
103.5	354.09	0.7838	0.7608
103.5	354.15	0.8142	0.7744
103.5	354.83	0.8993	0.8674
103.5	354.98	0.9194	0.8953
103.5	355.55	0.9674	0.9571
103.5	355.81	0.9859	0.9791

^aExpanded uncertainties (U), with a level of confidence of 95%: U (pressure): 0.51 kPa; U (temperature): 0.14 K; U (liquid mole fraction referred to isopropanol): 0.0168 mol/mol; U (vapor mole fraction referred to isopropanol): 0.0184 mol/mol.

of chemical engineering, and the engineering design of processes requires equipment that takes into account the thermodynamic and hydrodynamic characteristics of the system to be treated. For this reason, a comparison with the thermodynamic models Predictive Soave–Redlich–Kwong (PSRK),³² Nonrandom Two-Liquid (NRTL),³³ UNIQUAC functional-group activity coefficients (UNIFAC),³⁴ and Wilson³⁵ has been also included in this paper.

2. EXPERIMENTAL SETUP

2.1. Chemicals. Demineralized water (ISO 3696 Q3, ASTM D 1193 TYPE 4) was used and isopropanol (CAS 67–63–0) was supplied by Sigma-Aldrich, with a purity of at least 99.8%, molecular weight of 60.10 g/mol and specific gravity <0.7840. Nitrogen (>99.99%, mole), coming from a tank storing N₂ in the liquid phase, is used for pressurizing the VLLE unit (Fisher Labodest VLLE 602) up to the desired pressure and as a carrier for the gas chromatograph. The characteristics of the chemical compounds employed in this work are summarized in Table 3.

2.2. Apparatus. The Process Thermodynamics laboratory (PT lab) of Politecnico di Milano is located at Dipartimento di Chimica, Materiali e Ingegneria Chimica “Giulio Natta” and has been created within the “Ingegneria Chimica – Energia (ICE)” collaboration with the aim of collecting experimental data of

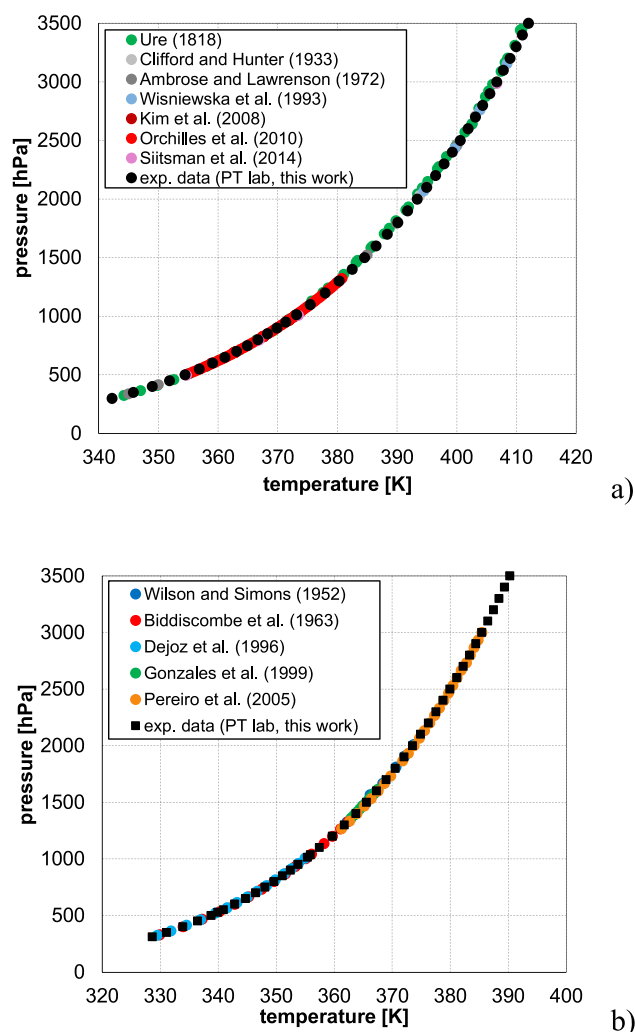


Figure 2. Vapor pressures of (a) water and (b) isopropanol obtained experimentally at the PT lab of Politecnico di Milano and from the literature.

phase equilibria of mixtures of interest for which no points can be found in the literature or in other existing databases.

The laboratory, which has been recently improved with a thermostatic bath for the study of liquid–liquid equilibria (LLE) at different temperatures, comprises a unit for vapor–liquid equilibria Fisher Labodest VLLE 602 (Figure 1) and a gas chromatograph Agilent 7820A, with a TCD detector.

The VLLE unit is an all-glass recirculating still that ensures that both liquid and vapor phases are in intimate contact during boiling and in contact with the temperature probe. The system can be used at atmospheric pressure, vacuum, and overpressure conditions up to 400 kPa. The equilibrium temperature is measured with a digital temperature logging module and a Pt 100 resistance thermometer. The accuracy is ± 0.1 K, and the resolution is 0.01 K, as indicated by the supplier. A precision pressure transmitter is used to measure and control the pressure in the equilibrium cell, with a pressure accuracy of ± 10 Pa as indicated by the supplier. This apparatus allows reaching pressures higher than atmospheric pressure (up to 400 kPa) by injection of high purity nitrogen, under pressure control of two manual valves and under automatic control.

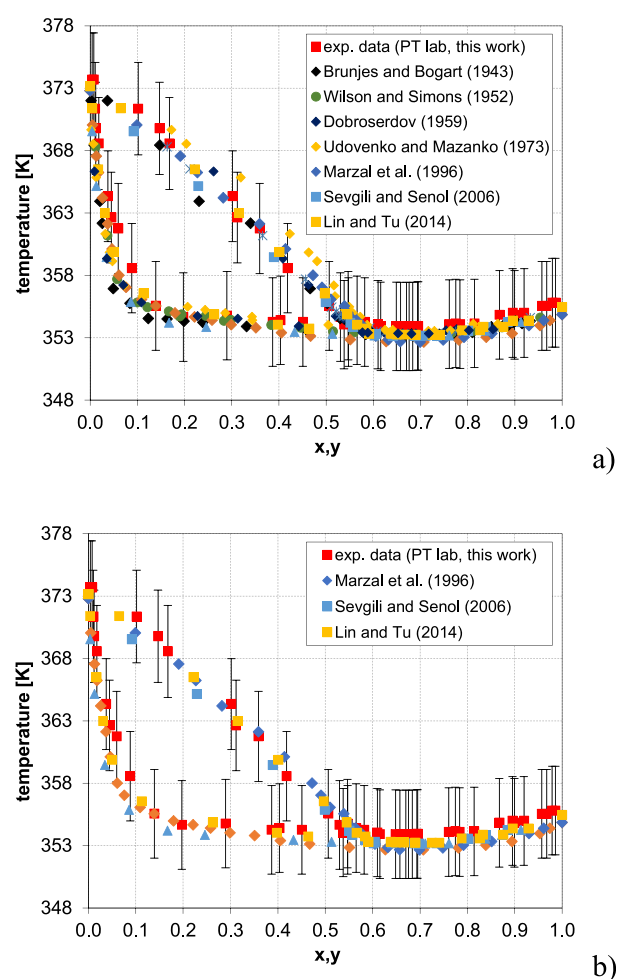


Figure 3. Experimental VLE data collected at the PT lab of Politecnico di Milano at 103.5 kPa and from literature sources for the system isopropanol–water at 101.3 kPa considering (a) most of all the literature sources and (b) a selection of the most recent available literature sources. The error bars represent an error of 1%.

The GC Agilent 7820A is provided with a column HP-INNOWax 30 m long, with an inner diameter of 0.25 mm and a film thickness of 0.5 μm .

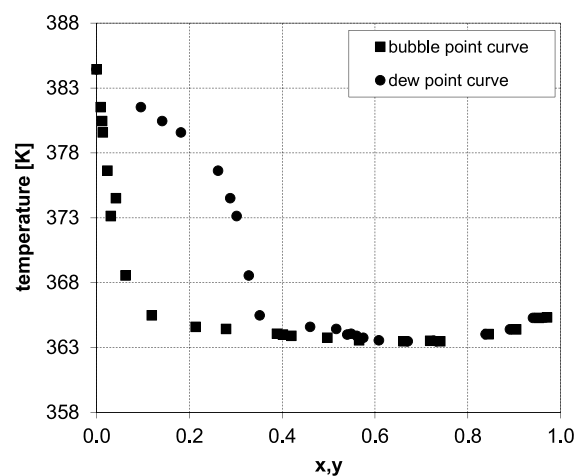
3. EXPERIMENTAL PROCEDURE AND VALIDATION

3.1. Experimental Procedure. The experimental procedure employed for the work presented in this paper is divided into four steps:

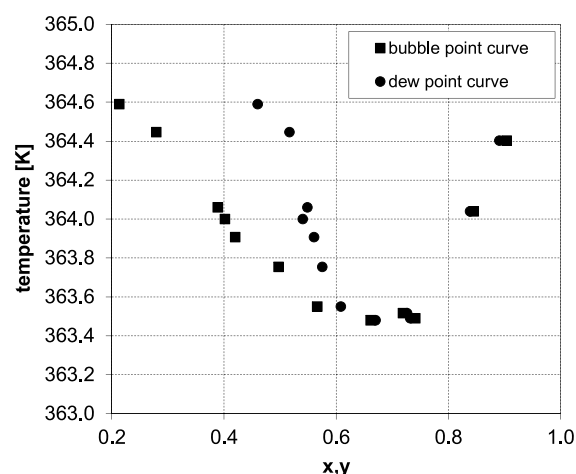
1. preparation and feeding
2. setting the operating conditions and reaching equilibrium
3. sampling
4. emptying the unit

Preparation and Feeding. A check that the ventilation valves, the PTFE valves for pressure balance in the side branches, and the sampling valves are open is made, then the mixture to be fed is prepared under the hood in the beaker (100 mL) and it is fed to the feed tank of the VLLE unit. The heater is then filled in, so that the electric heating element is completely submerged.

Setting the Operating Conditions and Reaching Equilibrium. The desired pressure value is set, the ventilation valves are closed, and the opening of the lateral throttle valves (one to increase and the other one to lower the pressure) is set until the



a)



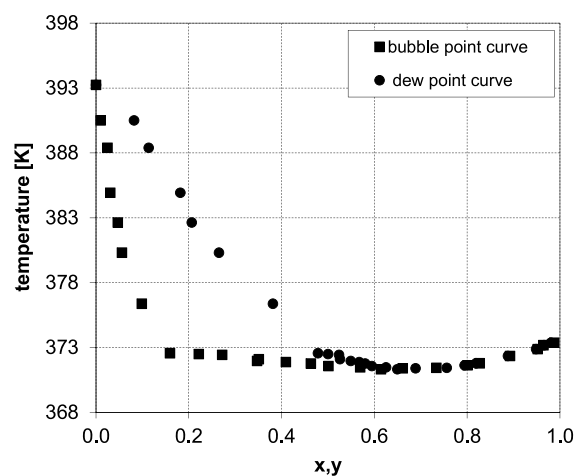
b)

Figure 4. Experimental VLE data collected at the PT lab of Politecnico di Milano for the isopropanol–water system at 150 kPa (a) for a mole fraction of isopropanol ranging from 0 to 1 and (b) detail for the composition range close to the azeotropic point.

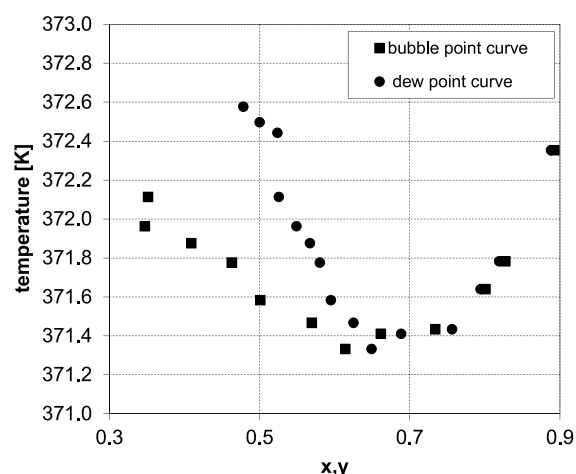
set point pressure is reached. The magnetic stirrer is activated, and the thermal power to be supplied to heat up the mixture is provided, then the mixture is allowed to boil. When the temperature of the vapors reaching the top of the equilibrium chamber remains constant for at least 10 min (a maximum oscillation of $\pm 0.5\%$ is tolerated for VLE of mixtures) and the formation of 1–2 drops of condensate per second is observed, equilibrium is considered to be reached.

Sampling. The gas chromatograph is switched to the analysis mode, and the analysis of the composition of the liquid phase and of the vapor phase is performed. The VLLE unit at the PT lab is in-line with the gas chromatograph, so that the vapor phase can be analyzed directly in line (in case the carrier is not present as component of the mixture to be analyzed) or, as done in this work, in its condensate phase. The gas and the liquid analyses are repeated at least three times making sure that the temperature at which the equilibrium is evaluated is always the same, to guarantee the repeatability of the experiment.

Emptying the Unit. At the end of the experimental procedure the unit is emptied and carefully cleaned to make it ready for further measurements.



a)



b)

Figure 5. Experimental VLE data collected at the PT lab of Politecnico di Milano for the isopropanol–water system at 200 kPa (a) for a mole fraction of isopropanol ranging from 0 to 1 and (b) detail for the composition range close to the azeotropic point.

3.2. Validation of the Experimental Procedure. To validate the unit at the PT lab, the data of water and IPA vapor pressures (Table 4 and Table 5) and VLE data for the IPA–water binary mixture at a pressure of about 101.3 kPa (103.5 kPa) have been collected (Table 6). These data were compared with the data available in the literature^{23,36–46} under operating conditions compatible with the operation of the VLLE unit present in the PT lab for vapor pressure estimation (Figure 2) and with the available experimental data at 101.3 kPa for the binary mixture VLE (Figure 3). In Figure 3a the error bars representing an error of 1% are also reported. In addition, because of the high number of sources available in the literature at 101.3 kPa, in Figure 3b a selection has been made so that the plot is more clearly understandable to the reader.

Both for the vapor pressure and for the VLE of the binary mixture at about 101.3 kPa a good agreement is obtained, confirming that the experimental procedure and the collection of points at different temperatures and compositions can be considered reliable. The points collected in this work are reported in Table 6.

In addition, the test on the thermodynamic consistency based on the methodology proposed by Redlich and Kister⁴⁷ was

performed for all the obtained experimental data. This test was chosen to verify the thermodynamic consistency of the experimental data because of its wide use in the literature, for instance in the works by Martin et al.,⁴⁸ Vargas et al.,⁴⁹ Parsana et al.⁵⁰ and Das et al.⁵¹ The area test has been chosen also because it does not require (and the results do not depend on) any thermodynamic model, differently from a point-to-point test.

4. RESULTS

Results for the different isobaric vapor–liquid equilibrium curves are reported in Figure 4 at 150 kPa, in Figure 5 at 200 kPa,

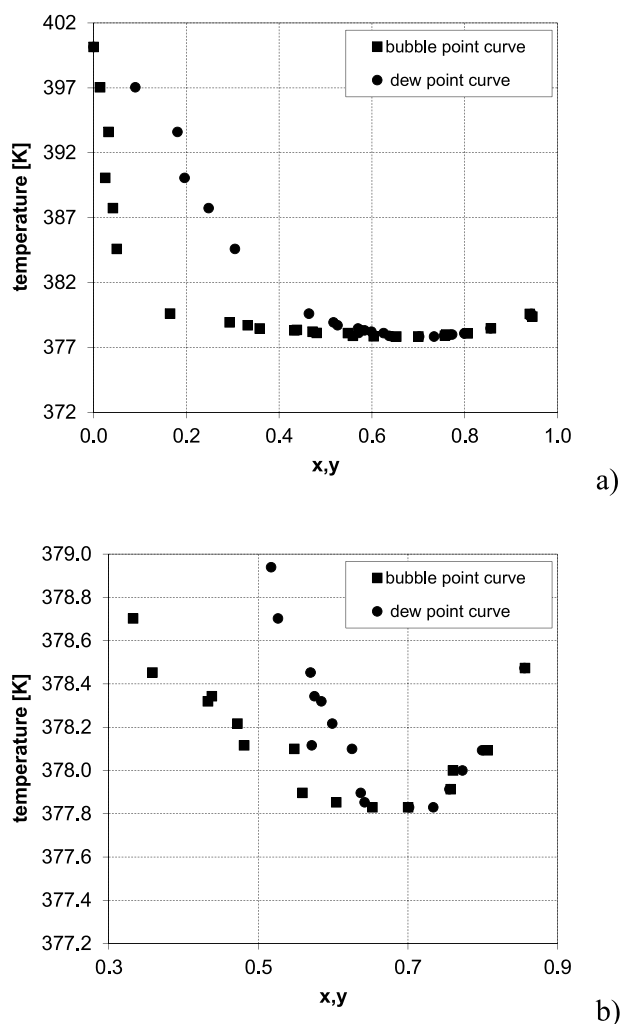


Figure 6. Experimental VLE data collected at the PT lab of Politecnico di Milano for the isopropanol–water system at 250 kPa (a) for a mole fraction of isopropanol ranging from 0 to 1 and (b) detail for the composition range close to the azeotropic point.

in Figure 6 at 250 kPa, and in Figure 7 at 300 kPa. The resulting experimental points are also reported in Tables 6 to Table 10, together with the estimated uncertainties for temperature, pressure, and composition of the vapor and of the liquid phases (the average of each uncertainty has been calculated excluding the points for the pure components, for which the uncertainty in the mole fractions is 0).

The provided uncertainties are expanded uncertainties: they have been estimated considering a coverage factor equal to 2,

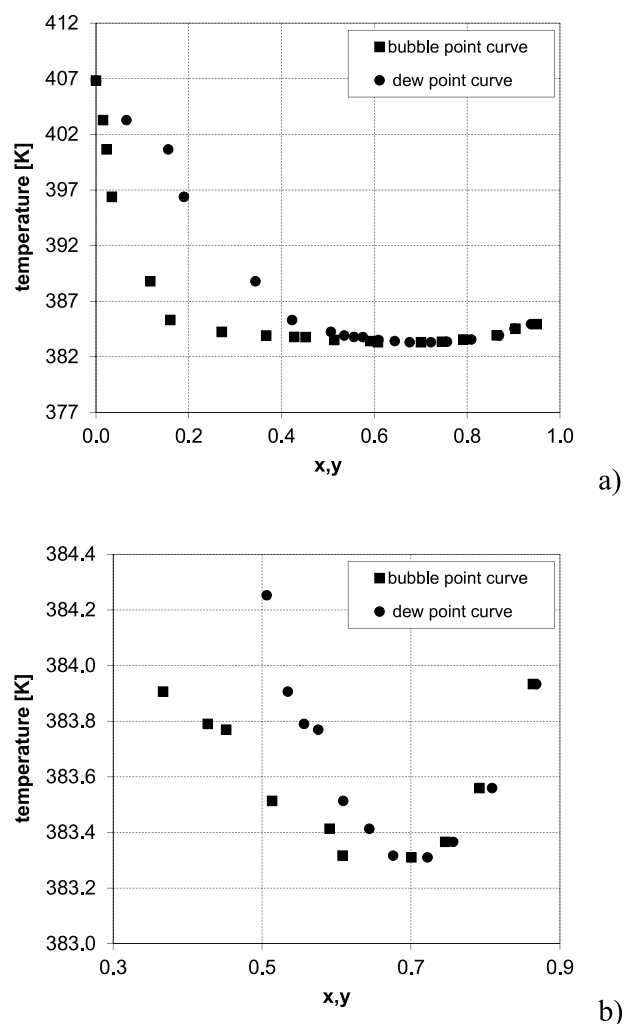


Figure 7. Experimental VLE data collected at the PT lab of Politecnico di Milano for the isopropanol–water system at 300 kPa (a) for a mole fraction of isopropanol ranging from 0 to 1 and (b) detail for the composition range close to the azeotropic point.

which gives an appropriate 95% level of confidence for the interval.

The method adopted in this work for evaluating and expressing the uncertainty of measurement results follows the policy instituted at NIST (National Institute of Standards and Technology) in 1992. This policy is based on the approach recommended by the CIPM (International Committee for Weights and Measures) in 1981.⁵²

As a result of the thermodynamic consistency test, all data satisfy the criteria. A default area tolerance equal to 10% has been selected as the threshold value to define whether each experimental data set was able to successfully pass the test. The numerical results of the consistency test are available in Table 11.

A minimum temperature azeotrope occurs for all the considered pressures, confirming the nonideality of the mixture. On the right side of the azeotrope, for high mole fractions of isopropanol, the bubble point curve becomes very close to the dew point curve, differently from the trend at low isopropanol contents. These trends are confirmed also by comparison with the isobaric curve by Wilson and Simons²³ at about 400 kPa

Table 7. Data of Temperature, Pressure, Composition in the Liquid and in the Vapor Phases Collected at the PT lab of Politecnico di Milano at a Pressure of 150 kPa and Related Average Uncertainties (x is the Mole Fraction in the Liquid Phase and y is the Mole Fraction in the Vapor Phase of the Most Volatile Component, i.e., Isopropanol)^a

pressure [kPa]	temp [K]	x	y
150	381.53	0.0093	0.0953
150	380.46	0.0120	0.1414
150	379.58	0.0137	0.1817
150	376.62	0.0233	0.2617
150	373.13	0.0306	0.3021
150	374.51	0.0415	0.2878
150	368.55	0.0625	0.3280
150	365.48	0.1193	0.3516
150	364.59	0.2134	0.4602
150	364.45	0.2790	0.5167
150	364.06	0.3888	0.5489
150	364.00	0.4017	0.5403
150	363.91	0.4199	0.5606
150	363.75	0.4972	0.5750
150	363.55	0.5662	0.6084
150	363.48	0.6614	0.6705
150	363.52	0.7191	0.7263
150	363.49	0.7408	0.7320
150	364.04	0.8451	0.8385
150	364.40	0.9040	0.8910
150	365.29	0.9534	0.9406
150	365.31	0.9709	0.9675

^aExpanded uncertainties (U), with a level of confidence of 95%: U (pressure): 0.36 kPa; U (temperature): 0.19 K; U (liquid mole fraction referred to isopropanol): 0.0174 mol/mol; U (vapor mole fraction referred to isopropanol): 0.0178 mol/mol.

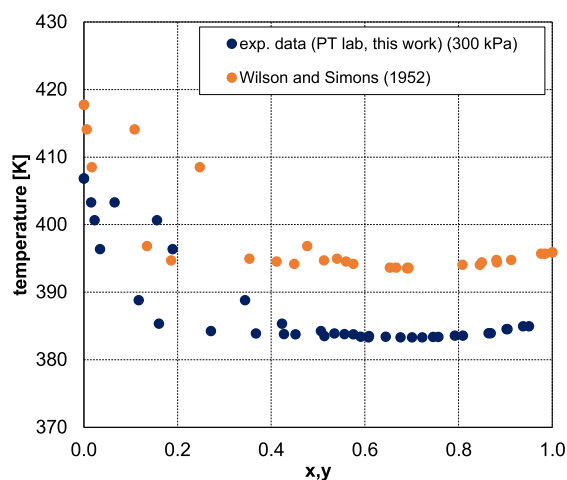


Figure 8. Comparison of the experimental VLE data collected at the PT lab of Politecnico di Milano for the isopropanol–water system at 300 kPa and the ones by Wilson and Simons²³ at about 400 kPa.

(Figure 8), in addition to the ones available in the literature at 101.3 kPa (Figure 3).

Figure 9 shows a comparison of the obtained curves at different pressures, which highlights the variation of temperature levels as expected with increasing pressure and of the azeotropic point, that slightly moves toward a higher mole fraction of isopropanol as the pressure increases.

Table 8. Data of Temperature, Pressure, Composition in the Liquid and in the Vapor Phases Collected at the PT lab of Politecnico di Milano at a Pressure of 200 kPa and Related Average Uncertainties (x is the Mole Fraction in the Liquid Phase and y is the Mole Fraction in the Vapor Phase of the Most Volatile Component, i.e., Isopropanol)^a

pressure [kPa]	temp [K]	x	y
200	390.50	0.0103	0.0820
200	388.40	0.0249	0.1135
200	384.93	0.0308	0.1819
200	382.64	0.0470	0.2067
200	380.32	0.0559	0.2654
200	376.37	0.0988	0.3811
200	372.58	0.1596	0.4784
200	372.50	0.2219	0.5001
200	372.44	0.2724	0.5239
200	371.96	0.3474	0.5493
200	372.11	0.3514	0.5259
200	371.88	0.4092	0.5673
200	371.78	0.4632	0.5804
200	371.58	0.5011	0.5948
200	371.47	0.5695	0.6252
200	371.33	0.6143	0.6493
200	371.41	0.6614	0.6887
200	371.43	0.7339	0.7563
200	371.64	0.8009	0.7943
200	371.78	0.8273	0.8191
200	372.35	0.8926	0.8880
200	372.88	0.9521	0.9488
200	373.18	0.9645	0.9659
200	373.38	0.9877	0.9807

^aExpanded uncertainties (U), with a level of confidence of 95%: U (pressure): 0.80 kPa; U (temperature): 0.11 K; U (liquid mole fraction referred to isopropanol): 0.0169 mol/mol; U (vapor mole fraction referred to isopropanol): 0.0159 mol/mol.

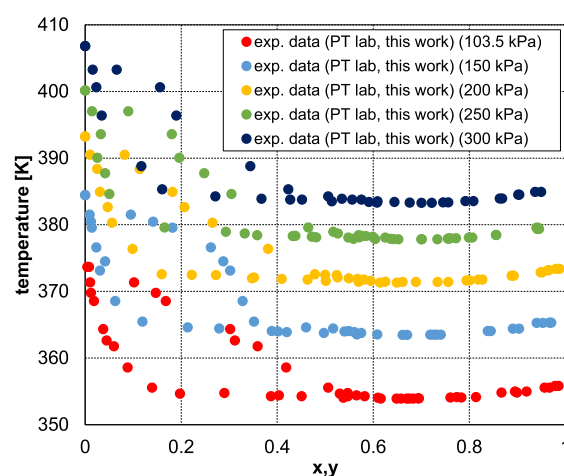


Figure 9. Comparison of the VLE experimental data collected at the PT lab of Politecnico di Milano for the isopropanol–water system at different pressures.

5. THERMODYNAMIC MODELING

The collected experimental data have been compared with the results obtained with different thermodynamic models, commonly employed to describe strongly nonideal mixtures as the one studied in this work. Methods based both on a ϕ/ϕ

Table 9. Data of Temperature, Pressure, Composition in the Liquid and in the Vapor Phases Collected at the PT lab of Politecnico di Milano at a Pressure of 250 kPa and Related Average Uncertainties (x Is the Mole Fraction in the Liquid Phase and y Is the Mole Fraction in the Vapor Phase of the Most Volatile Component, i.e., Isopropanol)^a

pressure [kPa]	temp [K]	x	y
250	397.06	0.0141	0.0898
250	390.06	0.0255	0.1961
250	393.61	0.0324	0.1807
250	387.73	0.0417	0.2481
250	384.60	0.0503	0.3046
250	379.60	0.1652	0.4644
250	378.94	0.2932	0.5170
250	378.70	0.3323	0.5262
250	378.45	0.3583	0.5698
250	378.32	0.4326	0.5842
250	378.34	0.4379	0.5750
250	378.22	0.4722	0.5990
250	378.12	0.4810	0.5712
250	378.10	0.5481	0.6252
250	377.90	0.5590	0.6369
250	377.85	0.6040	0.6420
250	377.83	0.6525	0.7016
250	377.83	0.7001	0.7338
250	377.91	0.7573	0.7554
250	378.00	0.7602	0.7727
250	378.09	0.8065	0.7991
250	378.47	0.8563	0.8558
250	379.58	0.9400	0.9435
250	379.38	0.9456	0.9418

^aExpanded uncertainties (U), with a level of confidence of 95%: U (pressure): 0.93 kPa; U (temperature): 0.12 K; U (liquid mole fraction referred to isopropanol): 0.0158 mol/mol; U (vapor mole fraction referred to isopropanol): 0.0170 mol/mol.

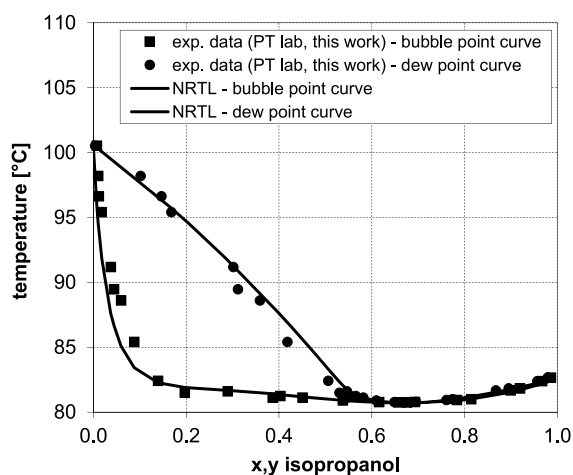


Figure 10. Comparison of the VLE experimental data collected at the PT lab of Politecnico di Milano for the isopropanol–water system at 103.5 kPa and the results obtained with the NRTL model with ASPEN Plus V11.0 default parameters.

approach, as PSRK, and on a γ/ϕ approach, as NRTL, UNIFAC, and Wilson, have been considered.

In this preliminary analysis of thermodynamic models, ASPEN Plus V11.0⁵³ has been used, with default parameters.

Table 10. Data of Temperature, Pressure, Composition in the Liquid and in the Vapor Phases Collected at the PT lab of Politecnico di Milano at a Pressure of 300 kPa and Related Average Uncertainties (x Is the Mole Fraction in the Liquid Phase and y Is the Mole Fraction in the Vapor Phase of the Most Volatile Component, i.e., Isopropanol)^a

pressure [kPa]	temp [K]	x	y
300	403.28	0.0156	0.0657
300	400.65	0.0231	0.1559
300	396.39	0.0344	0.1896
300	388.80	0.1171	0.3440
300	385.32	0.1603	0.4229
300	384.25	0.2712	0.5062
300	383.91	0.3671	0.5347
300	383.79	0.4271	0.5563
300	383.77	0.4518	0.5754
300	383.51	0.5137	0.6089
300	383.41	0.5909	0.6442
300	383.32	0.6079	0.6761
300	383.31	0.7007	0.7222
300	383.37	0.7457	0.7565
300	383.56	0.7920	0.8089
300	383.93	0.8637	0.8684
300	384.53	0.9043	0.9022
300	384.95	0.9501	0.9376

^aExpanded uncertainties (U), with a level of confidence of 95%: U (pressure): 0.51 kPa; U (temperature): 0.16 K; U (liquid mole fraction referred to isopropanol): 0.0186 mol/mol; U (vapor mole fraction referred to isopropanol): 0.0238 mol/mol.

Table 11. Results of the Redlich–Kister Area Thermodynamic Consistency Check Used to Assess the Thermodynamic Consistency of the VLE Data Collected at the PT lab of Politecnico di Milano at Pressures of 150 kPa, 200 kPa, 250 kPa, and 300 kPa for the Isopropanol–Water System

pressure [kPa]	Redlich–Kister area test result	consistency requirement
150	6.563	<10
200	0.097	<10
250	8.664	<10
300	5.996	<10

Table 12. %AAD of the Temperature Estimated with PSRK, NRTL, UNIFAC, and Wilson Models for the Isobaric Datasets Collected at the PT lab of Politecnico di Milano

pressure [kPa]	PSRK	NRTL	UNIFAC	WILSON
103.5	0.36%	0.31%	0.41%	0.40%
150	0.29%	0.26%	0.32%	0.36%
200	0.35%	0.35%	0.38%	0.43%
250	0.28%	0.27%	0.30%	0.39%
300	0.32%	0.37%	0.38%	0.47%

Regression of new thermodynamic parameters for the considered models is out of the scope of this work. Tables 12 to Table 14 report the results obtained with the different models as percent absolute average deviation (%AAD), defined as in eq 1, for the collected isobaric data at all the pressures (from 103.5 to 300 kPa), considering the calculated temperature and composition in the vapor phase at given pressure and composition in the liquid phase.

Table 13. %AAD of the Mole Fraction in the Vapor Phase of Isopropanol Estimated with PSRK, NRTL, UNIFAC and Wilson Models for the Isobaric Datasets Collected at the PT lab of Politecnico di Milano (for the set at 103.5 kPa the First Point Close to Pure Water Composition Has Not Been Considered)

pressure [kPa]	PSRK	NRTL	UNIFAC	WILSON
103.5	12.41%	14.95%	17.37%	16.74%
150	12.42%	19.56%	20.10%	22.56%
200	21.78%	31.21%	29.52%	32.25%
250	15.38%	24.78%	22.23%	24.93%
300	23.72%	37.70%	33.27%	39.08%

Table 14. %AAD of the Mole Fraction in the Vapor Phase of Water Estimated with PSRK, NRTL, UNIFAC, and Wilson Models for the Isobaric Datasets Collected at the PT lab of Politecnico di Milano

pressure [kPa]	PSRK	NRTL	UNIFAC	WILSON
103.5	5.75%	6.24%	7.65%	9.05%
150	7.18%	10.23%	11.72%	14.46%
200	9.80%	13.90%	14.38%	17.34%
250	9.06%	13.72%	13.59%	15.33%
300	9.18%	14.53%	14.63%	17.22%

$$\% \text{ AAD} = \frac{\sum_{i=1}^n \left| \frac{(\text{calc value}_i - \text{exp value}_i)}{\text{exp value}_i} \right|}{n} \quad (1)$$

where n is the number of data for each isobaric curve.

The comparison of the collected data and the values calculated with the NRTL model at 103.5 kPa is reported in Figure 10 (for clarity in reading only one method is reported). On the basis of the results reported from Tables 12 to Table 14 and in Figure 10, an acceptable representation of the system is obtained, also taking into account that isobaric data from 150 to 300 kPa were not available and were not used for the determination of key parameters for the thermodynamic models.

6. CONCLUSIONS

This work presents isobaric experimental vapor–liquid equilibrium data for the system isopropanol–water at 150 kPa, 200 kPa, 250 kPa, and 300 kPa to fill the gap existing in the literature. The equipment at the PT lab of Politecnico di Milano, consisting of an equilibrium unit and a gas-chromatograph, has been employed for this purpose, after validation with available experimental data from the literature. The collected experimental data were evaluated through the Redlich–Kister consistency test method. The system has been represented by considering thermodynamic models both based on a ϕ/ϕ approach and on a γ/ϕ approach, obtaining acceptable percent absolute average deviation values.

Future developments of this work could focus on the study of the vapor–liquid equilibrium curve at higher pressures up to 400 kPa or under vacuum to extend the range of conditions for which experimental data for this system are available and on the determination of new parameters for the selected thermodynamic models also based on the collected experimental data to better represent the isopropanol–water system.

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Notes

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